Prasad Sandip Joshi

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Education

Veermata Jijabai Technological Institute (VJTI) **Aug 2023 – May 2027**

Bachelor of Technology in Electronics and Telecommunication (Minor in DS & Machine Learning)

CGPA: 8.19/10 **MHT-CET:** 99.32 Percentile

Experience

Software Development Intern **June 2026 – July 2026**

Pidilite Industries Limited | Python, FastAPI, LangChain, Llama 3, Qdrant, PostgreSQL

- Contributed to **PilTantra+**, an enterprise **AI chatbot** enabling employees to retrieve accurate information from HR policies, SOPs, SharePoint resources, and internal knowledge bases using natural language.
- Developed and optimized a **Retrieval-Augmented Generation (RAG)** pipeline implementing document ingestion, intelligent chunking, vector indexing in **Qdrant**, semantic retrieval, and context-aware LLM response generation across enterprise documents and SOPs.
- Improved chatbot accuracy through **retrieval optimization**, **metadata filtering**, and **prompt engineering**, reducing hallucinations and improving answer relevance.
- Built and integrated **FastAPI** endpoints for document processing, retrieval, and chat interactions while collaborating cross-functionally to deliver a secure and scalable enterprise AI solution.

Projects

AI Financial Analyst | *Python, LangChain, Groq, FAISS, Pandas, RAG, Streamlit, Supabase* **June 2026**

- Developed an **agentic AI system** that autonomously analyzes financial CSV data and PDF annual reports through **natural language queries**, eliminating the need for manual formula writing or coding.
- Engineered a multi-tool **LangChain ReAct architecture** with autonomous tool selection across four components: a **Pandas agent** for code generation and execution, a **Plotly** chart generator, a FAISS-based **RAG** retrieval pipeline, and a direct LLM fallback.
- Built an end-to-end RAG pipeline using **PyPDFLoader**, **HuggingFace embeddings**,**OCR** support for scanned PDFs and **FAISS** for PDF question answering.
- Integrated **Supabase Authentication** with **persistent chat history** using **PostgreSQL**, enabling secure multi-user financial analysis.
- Deployed a Streamlit web application with **conversation memory**, **dynamic tool selection**, and interactive Plotly visualizations.

Credit Card Fraud Detection System | *Python, XGBoost, FastAPI, Streamlit, SHAP, Docker* **April 2026**

- Developed an end-to-end **fraud detection pipeline** using **SMOTE**, **PCA**, **Random Forest**, and **XGBoost** to classify highly imbalanced credit card transactions.
- Optimized the **XGBoost** model via **Optuna hyperparameter tuning** and threshold optimization, achieving **99.96% accuracy**, **89.32% precision**, **82.65% recall**, **85.86% F1-score**, and **98.26% ROC-AUC** on highly imbalanced transaction data.
- Built a **FastAPI REST API** and an interactive **Streamlit dashboard** supporting real-time and batch transaction predictions with **SHAP-based model** explainability.
- **Containerized** the application using **Docker** and implemented a basic **GitHub Actions CI/CD** pipeline to automate dependency installation, testing, and build verification while maintaining a modular and scalable project architecture.

NEP Timetable Management System | *Python, Genetic Algorithm, Node.js, JavaScript, MySQL* **October 2025**

- Designed and implemented a **Genetic Algorithm** to generate conflict-free academic timetables by optimizing faculty allocation, classroom assignment, and course scheduling under multiple constraints.
- Engineered the optimization pipeline using **selection**, **crossover**, **mutation**, and **fitness evaluation** to minimize scheduling conflicts and improve timetable quality.
- Developed a web-based management system with **role-based access** for administrators, faculty, and students using **Node.js**, **JavaScript**, and **MySQL**.

Bank Transaction Analytics & Reporting Platform | *PySpark, SQL, Qlik Sense, Python* **June 2026**

- Built a **PySpark** pipeline to clean, transform, and **govern** banking transaction data, improving data reliability for downstream reporting.
- Implemented **data quality checks** to identify missing values, duplicate transactions, invalid amounts, inconsistent categories, and data type errors.
- Used **SQL** to analyze transaction volumes, transaction values, customer segments, transaction types, failure rates, and monthly trends, translating raw data into **actionable business insights**.
- Developed an interactive **Qlik Sense** dashboard with KPI cards, trend analysis, and filters, delivering clear business reporting views to support **data-driven decision-making**.

Technical Skills

Languages:	Python, SQL, C++, JavaScript
Data Science & ML:	NumPy, Pandas, Scikit-learn, XGBoost, Random Forest, SHAP, PCA, SMOTE, Optuna
AI/GenAI:	LangChain, RAG, LLMs, Llama 3, FAISS, HuggingFace
Databases:	PostgreSQL, MySQL, SQLite, NoSQL, Qdrant
APIs & Deployment:	FastAPI, REST APIs, Streamlit, Docker
Cloud:	Oracle Cloud Infrastructure (OCI) – Fundamentals
Core Subjects:	Database Management Systems, Object Oriented Programming,Operating Systems,Data Structures
Developer Tools:	Git, GitHub, GitHub Actions, VS Code

Hackathon Experience

Smart India Hackathon (SIH) 2025

Team Lead

- Led a team of four in developing an AI-based **NEP Timetable Management System**, shortlisted among **100+ teams at the institute level** for the next round of Smart India Hackathon (SIH) 2025.